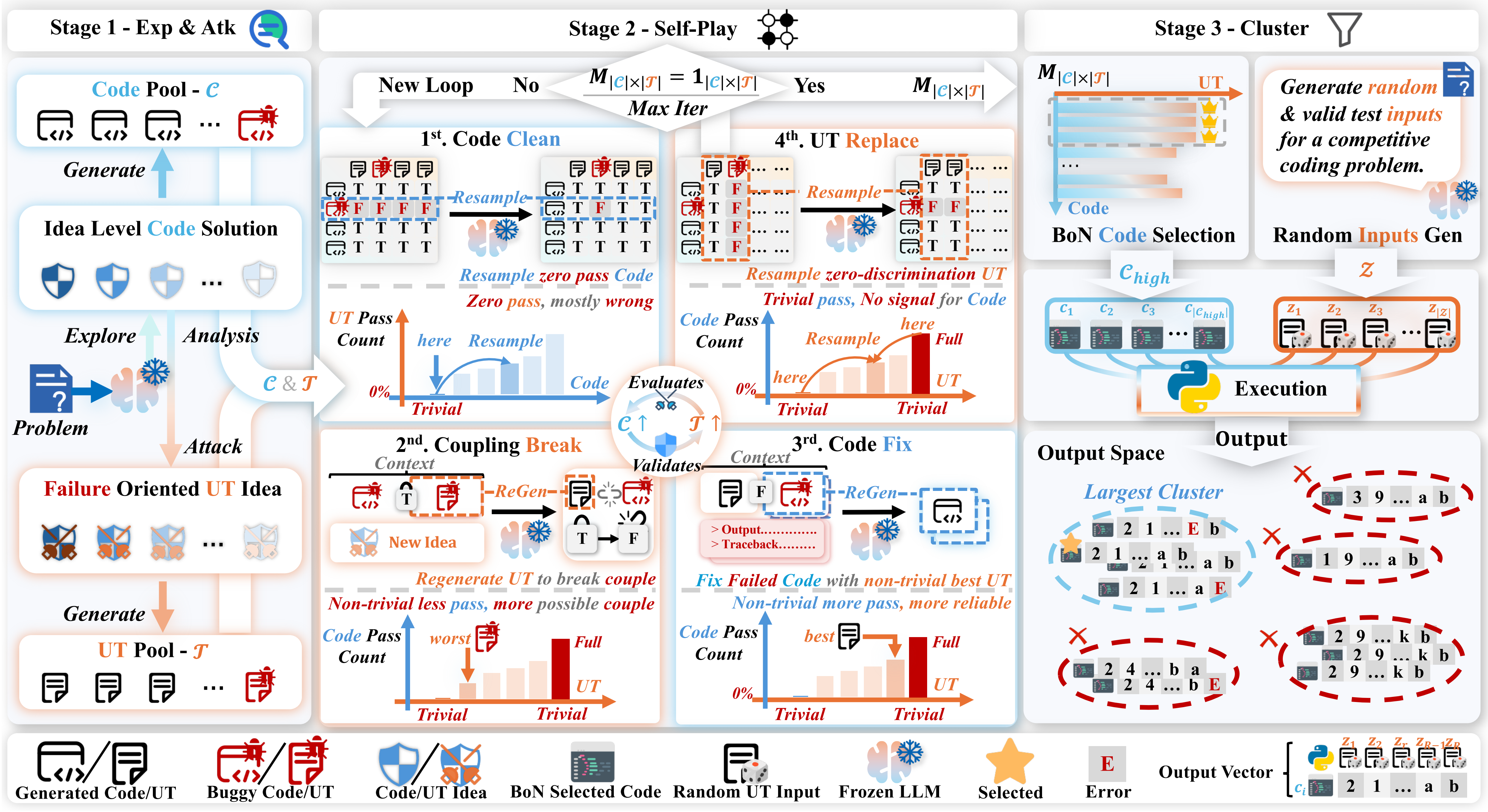
Stage 1 - Exp & Atk
Stage 2 - Self-Play
Stage 3 - Cluster
Code Pool - C
Generate
Idea Level Code Solution
Explore
Analysis
Problem
Attack
Failure Oriented UT Idea
Generate
UT Pool - T
C & T
New Loop
No
M|C|×|T| = 1|C|×|T|
Max Iter
Yes
M|C|×|T|
1st. Code Clean
Resample
Resample zero pass Code
Zero pass, mostly wrong
UT Pass Count
here
Resample
Code
0%
Trivial
4th. UT Replace
Resample
Resample zero-discrimination UT
Trivial pass, No signal for Code
Code Pass Count
Resample
here
Full UT
here
0%
Trivial
Trivial
Evaluates
C ↑
T ↑
Validates
2nd. Coupling Break
Context
ReGen
New Idea
Regenerate UT to break couple
Non-trivial less pass, more possible couple
Code Pass Count
worst
Full UT
Trivial
Trivial
3rd. Code Fix
Context
ReGen
> Output.............
> Traceback.........
Fix Failed Code with non-trivial best UT
Non-trivial more pass, more reliable
Code Pass Count
best
Full UT
0%
Trivial
Trivial
M|C|×|T|
UT
Code
BoN Code Selection
Generate random & valid test inputs for a competitive coding problem.
Random Inputs Gen
C_high
Z
Execution
Output
Output Space
Largest Cluster
Generated Code/UT
Buggy Code/UT
Code/UT Idea
BoN Selected Code
Random UT Input
Frozen LLM
Selected
Error
Output Vector